



School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067

Mid Semester Exam 2025-26

Program: MCA (II Semester)
Course: Computer Networks (CS408)
Time: 90 Minutes

Maximum Marks: 30
Exam Date: 16-03-2026
Student Roll No:

Note: Attempt any five questions, assume suitable data if missing.

- Q.1.** A bit stream **10011101** is transmitted using a Cyclic Redundancy Check (CRC) with generator polynomial $x^3 + 1$
- a) Generate the CRC bits for the given data. [2 Marks]
 - b) Write the complete transmitted codeword at the sender. [2 Marks]
 - c) Show the verification process at the receiver and state whether any error is detected or not. [2 Marks]
- Q.2.** Answer the following with proper explanation and examples:
- a) Write a note on round trip time (RTT) in networking. [2 Marks]
 - b) Discuss about the IP ranges of Class A, B, C and D. [2 Marks]
 - c) In a data link layer protocol, the following 8-bit character encodings are defined: A $\rightarrow$ 01000111 B $\rightarrow$ 11100011 FLAG $\rightarrow$ 01111110 ESC $\rightarrow$ 11100000
If character stuffing is used and the data field consists of the four-character sequence **A B ESC FLAG**, what exact bit sequence does the transmitter send on the link? [2 Marks]
- Q.3.** What is IP addressing? How is subnet addressing performed? The IP network **200.198.160.0** is using subnet mask **255.255.255.224**. Draw the subnets. [2+2+2= 6 Marks]
- Q.4.** Consider a **100 Mbps** link between an earth station (sender) and a satellite (receiver) at an altitude of **2100 km**. The signal propagates at a speed of 3×10^8 m/s. What is the time taken (in milliseconds) for the receiver to completely receive a **1000-byte** packet transmitted by the sender? [6 Marks]
- Q.5.** What is the need for protocol architecture? Explain the OSI model. [2+4 = 6 Marks]
- Q.6.** Differentiate between circuit switching and packet switching. Explain their working with suitable examples. [6 Marks]
- Q.7.** Write notes on any two of the following, with proper explanation and examples:
- a) Stop and Wait ARQ [3 Marks]
 - b) Selective Repeat ARQ [3 Marks]
 - c) Go Back N-ARQ [3 Marks]
- Q.8.** Explain Different Topologies with Suitable Diagrams. Also Differentiate Between Simplex, Half-Duplex, and Full-Duplex Transmission Modes with Examples. [4+2 = 6 Marks]



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End Semester Exam 2025-26

Program: MCA (II Semester)
Course: Computer Networks (CS-408)
Time: 10 AM to 1 PM

Maximum Marks: 50
Exam Date: 07/05/2026
Student Roll No:.....

Note: Attempt any four questions from Section A and any five questions from Section B. All questions from each section should be answered together in one place.

Section A

1. Explain the following with example: (2.5 + 2.5)
 - a) Persistent vs Non-Persistent HTTP
 - b) A 400-byte Internet Protocol packet, including a 20-byte header, is transmitted over a network with a Maximum Transmission Unit of 100 bytes. Explain the fragmentation process and calculate the number of fragments formed, the size of each fragment, and determine the values of the Don't Fragment, More Fragments, and fragment offset fields for each fragment.
2. Define framing in computer networks and explain the purpose of the Data Link Layer. (2.5 + 2.5)
3. Describe the following, with proper explanation and examples. (2.5 + 2.5): a) Count to Infinity problem, b) Silly Window Syndrome
4. Explain the complete lifecycle of a TCP connection from establishment to data transfer and termination. (5)
5. Given the following UDP datagrams, determine the source port number, destination port number, total length of the datagram, and length of the data. Also, identify whether the packet is sent from a client to a server or from a server to a client. (2.5 + 2.5): a) CB84000D001C001C, b) FFF00050FFFEFFFFE

Section B

1. Describe Distance Vector and Link State routing algorithms with comparison. (6)
2. Answer the following with proper explanations: (3 + 3)
 - a) Consider a TCP connection between a client and a server with the following specifications: the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t = 0$. Assume that there are no timeouts and errors during transmission. What is the size of the congestion window (in KB) at time $t + 60$ ms, after all acknowledgments have been processed?
 - b) Consider the three-way handshake mechanism followed during TCP connection establishment between hosts P and Q. Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends a TCP connection request message to Q with a TCP segment having SYN bit = 1, SEQ number = X, and ACK bit = 0. Suppose Q accepts the connection request. Identify and explain the information present in the TCP segment header sent from host Q to host P, specifically: SYN bit, Sequence Number (SEQ), ACK bit, Acknowledgment Number (ACK), and FIN bit.
3. Explain the following with examples (3 + 3): a) FTP, b) SMTP
4. Differentiate TCP and UDP in context of the header format. (6)
5. Explain CSMA/CD and CSMA/CA protocols with their working principles and key differences. (6)
6. For the following IP addresses (i) determine the class, (ii) address range of the class in dotted decimal notation, (iii) network mask (iv) indicate the network and host ids in dotted decimal form. (3 + 3)
 - a) 11000110001011010100100011100111, and b) 10111000010000100101110010000101